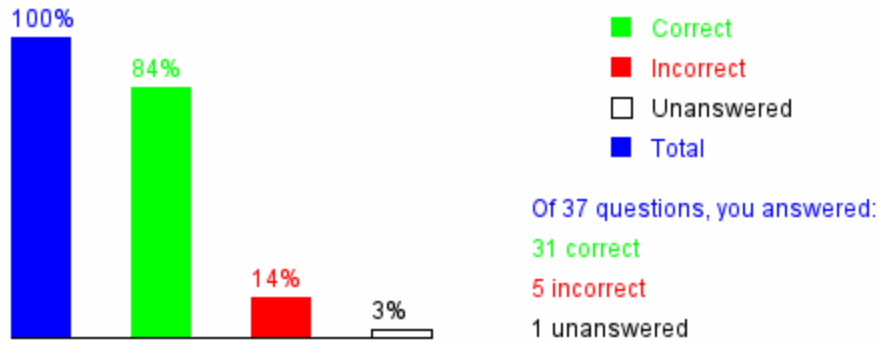


Introduction to Programming Using Python, Y. Daniel Liang

Many questions in this edition have been updated in the new edition. Please check with the publisher on the newest edition.

This quiz is for students to practice. A large number of additional quiz is available for instructors using Quiz Generator from the Instructor's Resource Website. Videos for Java, Python, and C++ can be found at <https://yongdanielliang.github.io/revelvideos.html>.

Chapter 10 Lists



Please send suggestions and errata to Dr. Liang at y.daniel.liang@gmail.com. Indicate which book and edition you are using.
 Thanks!

Section 10.2 List Basics

10.1 _____ creates a list.

- ☐ A. `list1 = list()`
- ☒ B. `list1 = []`
- ☒ C. `list1 = list([12, 4, 4])`
- ☒ D. `list1 = [12, 4, 4]`
- ☒ E. `list1 = [1, "3", "red"]`

Your answer BCDE is incorrect 
 Click [here](#) to show the correct answer

10.2 What is `list("abcd")`?

- ☒ A. `['a', 'b', 'c', 'd']`
- ☐ B. `['ab']`
- ☐ C. `['cd']`
- ☐ D. `['abcd']`

Your answer is correct 

10.3 Suppose `list1` is `[3, 4, 5, 20, 5, 25, 1, 3]`, what is `len(list1)`?

- ☐ A. 6
- ☐ B. 7
- ☒ C. 8
- ☐ D. 5
- ☐ E. 4

Your answer is correct 

10.4 Suppose `list1` is `[3, 4, 5, 20, 5, 25, 1, 3]`, what is `max(list1)`?

- ☐ A. 5
- ☐ B. 4
- ☐ C. 8
- ☒ D. 25
- ☐ E. 1

Your answer is correct



10.5 Suppose list1 is [3, 4, 5, 20, 5, 25, 1, 3], what is min(list1)?

- ☐ A. 5
- ☐ B. 4
- ☐ C. 8
- ☐ D. 25
- ☒ E. 1

Your answer is correct



10.6 Suppose list1 is [1, 3, 2], what is sum(list1)?

- ☐ A. 5
- ☐ B. 4
- ☒ C. 6
- ☐ D. 2
- ☐ E. 1

Your answer is correct



10.7 To shuffle list1, use _____.

- ☐ A. list1.shuffle()
- ☐ B. shuffle(list1)
- ☒ C. random.shuffle(list1)
- ☐ D. random.shuffleList(list1)

Your answer is correct



10.8 Suppose list1 is [1, 3, 2, 4, 5, 2, 1, 0], Which of the following is correct?

- ☒ A. print(list1[0])
- ☒ B. print(list1[:2])
- ☒ C. print(list1[:-2])
- ☒ D. print(list1[4:6])

Your answer is correct



10.9 Suppose list1 is [1, 3, 2, 4, 5, 2, 1, 0], What is list1[-1]?

- ☐ A. 3
- ☐ B. 5
- ☐ C. 1
- ☒ D. 0

Your answer is correct



10.10 Suppose list1 is [1, 3, 2, 4, 5, 2, 1, 0], What is list1[:-1]?

- ☐ A. 0
- ☐ B. [1, 3, 2, 4, 5, 2, 1]
- ☒ C. [1, 3, 2, 4, 5, 2]
- ☐ D. [1, 3, 2, 4, 5, 2, 1, 0]

Your answer C is incorrect



[Click here to show the correct answer](#)

10.11 Suppose list1 is [1, 3, 2], What is list1 * 2?

- ☐ A. [2, 6, 4]
- ☐ B. [1, 3, 2, 1, 3]
- ☒ C. [1, 3, 2, 1, 3, 2]
- ☐ D. [1, 3, 2, 3, 2, 1]

Your answer is correct



10.12 Suppose list1 = [0.5 * x for x in range(0, 4)], list1 is _____

- ☐ A. [0, 1, 2, 3]
- ☐ B. [0, 1, 2, 3, 4]
- ☒ C. [0.0, 0.5, 1.0, 1.5]
- ☐ D. [0.0, 0.5, 1.0, 1.5, 2.0]

Your answer is correct



10.13 list1 = [11, 2, 23] and list2 = [11, 2, 2], list1 < list2 is _____

- ☐ A. True
- ☒ B. False

Your answer is correct



10.14 list1 = [11, 2, 23] and list2 = [2, 11, 23], list1 == list2 is _____

- ☐ A. True
- ☒ B. False

Your answer is correct



10.15 To add 5 to the end of list1, use _____.

- ☐ A. list1.add(5)
- ☒ B. list1.append(5)
- ☐ C. list1.addLast(5)
- ☐ D. list1.addEnd(5)

Your answer is correct



10.16 To insert 5 to the third position in list1, use _____.

- ☐ A. list1.insert(3, 5)
- ☐ B. list1.insert(2, 5)
- ☒ C. list1.add(3, 5)

☐ D. `list1.append(3, 5)`

Your answer C is incorrect 
[Click here to show the correct answer](#)

10.17 To remove string "red" from list1, use _____.

- ☒ A. `list1.remove("red")`
☐ B. `list1.remove(red)`
☐ C. `list1.removeAll("red")`
☐ D. `list1.removeOne("red")`

Your answer is correct 

10.18 Suppose list1 is [3, 4, 5, 20, 5, 25, 1, 3], what is list1.index(5)?

- ☐ A. 0
☐ B. 4
☐ C. 1
☒ D. 2

Your answer is correct 

10.19 Suppose list1 is [3, 4, 5, 20, 5, 25, 1, 3], what is list1.count(5)?

- ☐ A. 0
☐ B. 4
☐ C. 1
☒ D. 2

Your answer is correct 

10.20 Suppose list1 is [3, 4, 5, 20, 5, 25, 1, 3], what is list1 after list1.sort()?

- ☐ A. [3, 4, 5, 20, 5, 25, 1, 3]
☒ B. [1, 3, 3, 4, 5, 5, 20, 25]
☐ C. [25, 20, 5, 5, 4, 3, 3, 1]
☐ D. [1, 3, 4, 5, 20, 5, 25, 3]

Your answer is correct 

10.21 Suppose list1 is [3, 4, 5, 20, 5, 25, 1, 3], what is list1 after list1.reverse()?

- ☐ A. [3, 4, 5, 20, 5, 25, 1, 3]
☐ B. [1, 3, 3, 4, 5, 5, 20, 25]
☐ C. [25, 20, 5, 5, 4, 3, 3, 1]
☐ D. [1, 3, 4, 5, 20, 5, 25, 3]
☒ E. [3, 1, 25, 5, 20, 5, 4, 3]

Your answer is correct 

10.22 Suppose list1 is [3, 4, 5, 20, 5, 25, 1, 3], what is list1 after list1.extend([34, 5])?

- ☒ A. [3, 4, 5, 20, 5, 25, 1, 3, 34, 5]
☐ B. [1, 3, 3, 4, 5, 5, 20, 25, 34, 5]

- ☐ C. [25, 20, 5, 5, 4, 3, 3, 1, 34, 5]
- ☐ D. [1, 3, 4, 5, 20, 5, 25, 3, 34, 5]
- ☐ E. [3, 1, 25, 5, 20, 5, 4, 3, 34, 5]

Your answer is correct



10.23 Suppose list1 is [3, 4, 5, 20, 5, 25, 1, 3], what is list1 after list1.pop(1)?

- ☐ A. [3, 4, 5, 20, 5, 25, 1, 3]
- ☐ B. [1, 3, 3, 4, 5, 5, 20, 25]
- ☒ C. [3, 5, 20, 5, 25, 1, 3]
- ☐ D. [1, 3, 4, 5, 20, 5, 25]
- ☐ E. [3, 1, 25, 5, 20, 5, 4]

Your answer is correct



10.24 Suppose list1 is [3, 4, 5, 20, 5, 25, 1, 3], what is list1 after list1.pop()?

- ☒ A. [3, 4, 5, 20, 5, 25, 1]
- ☐ B. [1, 3, 3, 4, 5, 5, 20, 25]
- ☐ C. [3, 5, 20, 5, 25, 1, 3]
- ☐ D. [1, 3, 4, 5, 20, 5, 25]
- ☐ E. [3, 1, 25, 5, 20, 5, 4]

Your answer is correct



10.25 "Welcome to Python".split() is _____

- ☒ A. ["Welcome", "to", "Python"]
- ☐ B. ("Welcome", "to", "Python")
- ☐ C. {"Welcome", "to", "Python"}
- ☐ D. "Welcome", "to", "Python"

Your answer is correct



10.26 What is list("a#b#c#d".split('#'))?

- ☒ A. ['a', 'b', 'c', 'd']
- ☐ B. ['a b c d']
- ☐ C. ['a#b#c#d']
- ☐ D. ['abcd']

Your answer is correct



10.27 What will be displayed by the following code?

```
myList = [1, 5, 5, 5, 5, 1]
max = myList[0]
indexOfMax = 0
for i in range(1, len(myList)):
    if myList[i] > max:
        max = myList[i]
        indexOfMax = i

print(indexOfMax)
```

- ☐ A. 0

- ☒ B. 1
- ☐ C. 2
- ☐ D. 3
- ☐ E. 4

Your answer is correct



10.28 What will be displayed by the following code?

```
myList = [1, 2, 3, 4, 5, 6]
for i in range(1, 6):
    myList[i - 1] = myList[i]

for i in range(0, 6):
    print(myList[i], end = " ")
```

- ☐ A. 2 3 4 5 6 1
- ☐ B. 6 1 2 3 4 5
- ☒ C. 2 3 4 5 6 6
- ☐ D. 1 1 2 3 4 5
- ☐ E. 2 3 4 5 6 1

Your answer is correct



Section 10.6 Copying Lists

10.29 What will be displayed by the following code?

```
list1 = [1, 3]
list2 = list1
list1[0] = 4
print(list2)
```

- ☐ A. [1, 3]
- ☒ B. [4, 3]
- ☐ C. [1, 4]
- ☐ D. [1, 3, 4]

Your answer is correct



Sections 10.7-10.8

10.30 What will be displayed by the following code?

```
def f(values):
    values[0] = 44

v = [1, 2, 3]
f(v)
print(v)
```

- ☐ A. [1, 44]
- ☐ B. [1, 2, 3, 44]
- ☒ C. [44, 2, 3]
- ☐ D. [1, 2, 3]

Your answer is correct



10.31 What will be displayed by the following code?

```
def f(value, values):  
    v = 1  
    values[0] = 44
```

```
t = 3  
v = [1, 2, 3]  
f(t, v)  
print(t, v[0])
```

- ☐ A. 1 1
- ☐ B. 1 44
- ☐ C. 3 1
- ☒ D. 3 44

Your answer is correct



10.32 What will be displayed by the following code?

```
def f(i, values = []):  
    values.append(i)  
    return values
```

```
f(1)  
f(2)  
v = f(3)  
print(v)
```

- ☐ A. [1] [2] [3]
- ☐ B. [1] [1, 2] [1, 2, 3]
- ☒ C. [1, 2, 3]
- ☐ D. 1 2 3

Your answer is correct



Section 10.10 Searching Lists

10.33 For the binarySearch function in Section 10.10.2, what is low and high after the first iteration of the while loop when invoking binarySearch([1, 4, 6, 8, 10, 15, 20], 11)?

- ☐ A. low is 0 and high is 6
- ☐ B. low is 0 and high is 3
- ☐ C. low is 3 and high is 6
- ☐ D. low is 4 and high is 6
- ☒ E. low is 0 and high is 5

Your answer E is incorrect



[Click here to show the correct answer](#)

10.35 If the binary search function returns -4, where should the key be inserted if you wish to insert the key into the list?

- ☒ A. at index 3
- ☐ B. at index 4
- ☐ C. at index 5
- ☐ D. at index 6

Your answer is correct



Section 10.11 Sorting Lists

10.36 Use the selectionSort function presented in this section to answer this question. Assume lst is [3.1, 3.1, 2.5, 6.4, 2.1], what is the content of list after the first iteration of the outer loop in the function?

- ☒ A. 3.1, 3.1, 2.5, 6.4, 2.1
- ☐ B. 2.5, 3.1, 3.1, 6.4, 2.1
- ☐ C. 2.1, 2.5, 3.1, 3.1, 6.4
- ☐ D. 3.1, 3.1, 2.5, 2.1, 6.4
- ☐ E. 2.1, 3.1, 2.5, 6.4, 3.1

Your answer A is incorrect



[Click here to show the correct answer](#)

10.37 Use the selectionSort function presented in this section to answer this question. What is list1 after executing the following statements?

```
list1 = [3.1, 3.1, 2.5, 6.4]
selectionSort(list1)
```

- ☐ A. list1 is 3.1, 3.1, 2.5, 6.4
- ☒ B. list1 is 2.5 3.1, 3.1, 6.4
- ☐ C. list1 is 6.4, 3.1, 3.1, 2.5
- ☐ D. list1 is 3.1, 2.5, 3.1, 6.4

Your answer is correct

